

BABEȘ-BOLYAI UNIVERSITY

FACULTY OF PHYSICS

COMPUTATIONAL PHYSICS

Master's thesis



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June 24, 2026

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**Design and Construction of a Low-Cost
Educational Spectrophotometer**



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Introduction and Objectives

Since vision is one of our most important and highly developed senses, it is natural that humans have long sought to build instruments capable of performing similar tasks in a quantitative and objective manner. Just as the human eye can often provide a qualitative impression of whether a sample of water is clean or contaminated, a spectrophotometer can analyze the interaction of light with matter and provide quantitative information about the composition and concentration of substances dissolved in a solution. This is widely used in chemistry, biology, medicine, and environmental monitoring. In particular, UV-visible (UV-Vis) absorption spectrophotometry is built on the Beer-Lambert law, which states that the amount of light absorbed by a solution is proportional to the concentration of the absorbing substance. This makes it an indispensable tool in both research and educational settings, where it underlies experiments ranging from the quantification of biomolecules to the study of reaction kinetics [1][2].

Despite its importance, access to absorption spectrophotometry is far from universal. Commercial UV-Vis spectrophotometers, while offering excellent performance, are expensive instruments that many schools and universities, particularly those with limited budgets, cannot afford [3]. Beyond the financial barrier, modern commercial instruments present a further challenge in educational contexts: they function as so-called "black boxes", hiding from the user the physical principles that underlie the measurement. Students who interact only with polished commercial devices are often left with little understanding of how a spectrum is actually generated, what role the diffraction grating plays, or why alignment and calibration matter [4][5]. This gap between instrument use and instrument understanding is a genuine obstacle to deep learning in physics and chemistry education.

In response to these challenges, a growing body of work over the past two decades has explored the construction of low-cost, home-built spectrophotometers for educational use. These range from very simple devices assembled from compact discs, flashlights, and cardboard [6], to more refined systems built from LEGO bricks [7][8], 3D-printed parts [4], or smartphone platforms [9]. A common finding in this literature is that building an instrument from scratch (selecting components, aligning optics, calibrating the wavelength scale, and troubleshooting) provides a qualitatively richer learning experience than simply operating a commercial device. Students who build their own spectrophotometers develop a hands-on understanding of diffraction, Beer-Lambert behavior,

and sources of experimental error that would otherwise remain abstract [5][8]. At the same time, these devices can bring quantitative spectroscopic measurement within financial reach of institutions that would otherwise have none [6].

The present thesis is situated within this context. The goal was to design, construct, and characterize a low-cost absorption spectrophotometer intended primarily for educational use. The instrument is developed with transparency in mind: every component is chosen and arranged so as to make the underlying physics as visible and understandable as possible to anyone assembling or using it. Alongside the hardware, a dedicated software system is developed to handle data acquisition, wavelength calibration, and the computation of absorbance spectra [4].

More specifically, the thesis begins by reviewing the theoretical foundations of UV-Vis absorption spectroscopy, including the Beer-Lambert law and the general operating principles of spectrophotometric instruments. It then surveys existing low-cost and DIY spectrophotometer designs from the literature, identifying key design choices and their trade-offs. The bulk of the work describes the hardware and software of the constructed instrument in detail, including the practical obstacles encountered and the solutions adopted. Finally, the performance of the device is validated through experimental measurements, and its limitations and potential improvements are discussed [10].

By addressing these objectives, this thesis aims to contribute to the broader effort of making spectroscopic measurement accessible, transparent, and educationally meaningful — an effort that is increasingly recognized as valuable not only in chemistry, but in physics education as well [4][5].

1 Theoretical Background

1.1 Spectroscopy in General

Spectroscopy is the branch of science concerned with the interaction between electromagnetic radiation and matter. When light encounters a substance, several things can happen: it may be transmitted, reflected, scattered, or absorbed, depending on the nature of the material and the wavelength of the radiation. Spectroscopy exploits these interactions to obtain information about the physical and chemical properties of substances [2]. Different regions of the electromagnetic spectrum give rise to different types of spectroscopic techniques. X-ray spectroscopy probes the inner electrons of atoms, infrared spectroscopy reveals molecular vibrations, and UV-visible spectroscopy, which is the focus of this work, involves transitions between electronic energy levels [1]. As shown in Figure 1.1, the visible spectrum occupies a narrow band of wavelengths, situated between the ultraviolet and infrared regions.

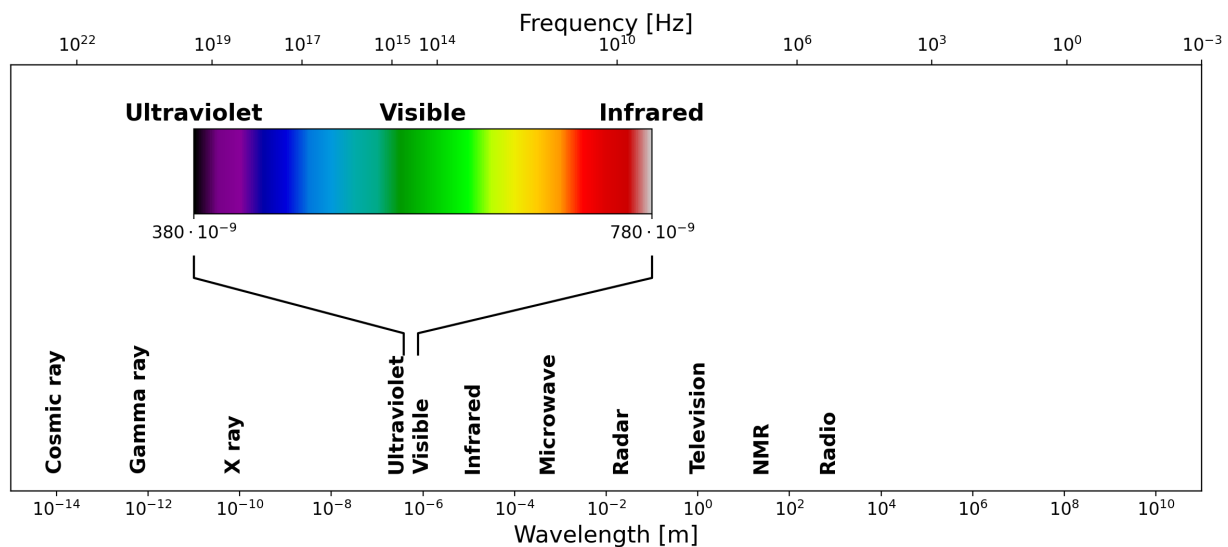


Figure 1.1: The electromagnetic spectrum, showing the full range of radiation from cosmic rays to radio waves. The visible region, spanning approximately 380 to 780 nm, is highlighted with its characteristic color gradient. Both wavelength and frequency scales are shown.

The electromagnetic spectrum spans an enormous range of wavelengths and frequencies, from very short gamma rays at one extreme to long radio waves at the other. The UV-visible region occupies only a small portion of this spectrum, roughly from 190 nm to 800 nm, but it is particularly

important because the energy of photons in this range corresponds to the energy differences between electronic states in many molecules. When a photon of the right energy strikes a molecule, it can be absorbed, promoting an electron from a lower energy level to a higher one. The wavelengths at which this absorption occurs are characteristic of the molecular structure, making UV-visible spectroscopy a powerful tool for both qualitative identification and quantitative analysis of chemical substances [1][2].

1.2 Absorption (UV/Vis) Spectroscopy

The quantitative foundation of absorption spectrophotometry is the Beer-Lambert law, which describes the relationship between the concentration of an absorbing substance and the amount of light it absorbs. To understand it, consider a beam of monochromatic light passing through a solution of an absorbing compound held in a transparent container called a cuvette [10]. Some of the light will be absorbed by the molecules in solution, and the rest will be transmitted through to the detector.

The transmittance T of the solution is defined as the ratio of the transmitted light intensity I to the incident light intensity I_0 :

$$T = \frac{I}{I_0} \quad (1)$$

Transmittance is often expressed as a percentage. However, for analytical purposes, it is more convenient to work with the absorbance A , defined as the negative logarithm of the transmittance:

$$A = -\log_{10}(T) = \log_{10}\left(\frac{I_0}{I}\right) \quad (2)$$

The Beer-Lambert law states that the absorbance is directly proportional to both the concentration of the absorbing species and the path length of the light through the sample [2]:

$$A = \varepsilon \cdot l \cdot c \quad (3)$$

where ε is the molar absorptivity (also called the molar extinction coefficient), a constant characteristic of the absorbing substance at a given wavelength, l represents the path length of light through the sample, as illustrated by the cuvette width in Figure 1.2 (1 cm in standard cuvettes), and c is the concentration of the absorbing species. The molar absorptivity has units of $\text{L mol}^{-1}\text{cm}^{-1}$ and can vary enormously between different compounds and different wavelengths [1].

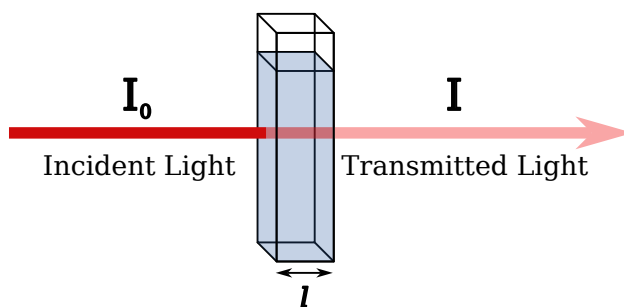


Figure 1.2: Schematic illustration of the Beer-Lambert law. A beam of incident light with intensity I_0 passes through a sample of path length l , emerging with reduced intensity I due to absorption by the medium.

The Beer-Lambert law is the cornerstone of quantitative spectrophotometry. If the molar absorptivity of a substance is known, measuring the absorbance of a solution immediately gives its concentration. In practice, a calibration curve is constructed by plotting the absorbance of standard solutions against their known concentrations [10], as shown in the right panel of Figure 1.3, allowing for the determination of unknown sample concentrations via linear interpolation. The law holds well at low to moderate concentrations, but deviations can occur at high concentrations, where interactions between molecules begin to affect the absorption behavior, or when the incident light is not truly monochromatic [1].

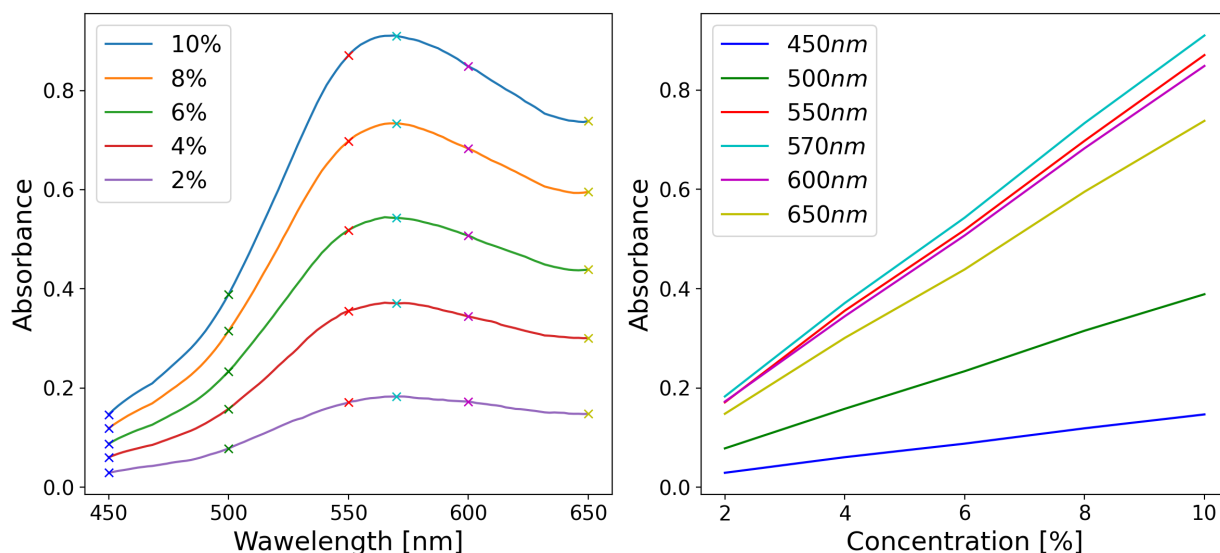


Figure 1.3: Left: example absorption spectra of a colored solution at five different concentrations, illustrating the characteristic shape of an absorption band and the increase in absorbance with concentration. Right: absorbance as a function of concentration at six specific wavelengths, illustrating the linear relationship predicted by the Beer-Lambert law.

1.2.1 Absorption Spectra and Molecular Structure

When the absorbance of a substance is measured not just at a single wavelength but across a range of wavelengths, the result is an absorption spectrum (a plot of absorbance as a function of wavelength). Each substance has a characteristic absorption spectrum that reflects its molecular structure, specifically the energies of its electronic transitions. The wavelength at which the absorbance reaches its maximum is called λ_{max} , and quantitative measurements are most sensitive at this wavelength, since small changes in concentration cause the largest changes in absorbance [1][10].

The shape and position of absorption bands in the UV-visible spectrum depend on the electronic structure of the molecule. In organic molecules, the electrons most likely to be involved in UV-visible absorption are the π electrons in conjugated systems and the non-bonding lone pair electrons on heteroatoms such as oxygen and nitrogen. As the degree of conjugation in a molecule increases, the energy gap between the ground state and the excited state decreases, and the absorption band shifts to longer wavelengths: a phenomenon known as a bathochromic or red shift. This is why, for example, beta-carotene, which has a long conjugated chain, absorbs strongly in the visible region and appears orange, while simpler molecules with fewer conjugated bonds absorb in the UV and are colorless [1][10].

1.2.2 Structure and Operation of Absorption Spectrometers

A UV-visible absorption spectrophotometer is an instrument designed to measure the absorbance of a sample as a function of wavelength. Despite the many variations in design and complexity that exist, all spectrophotometers share the same fundamental architecture, consisting of five basic components: a light source, a wavelength selector, a sample holder, a detector, and a readout system [1]. The spatial arrangement and theoretical operation of the primary optical components are schematically represented in Figure 1.4, with the readout system typically integrated into a modern computer interface.

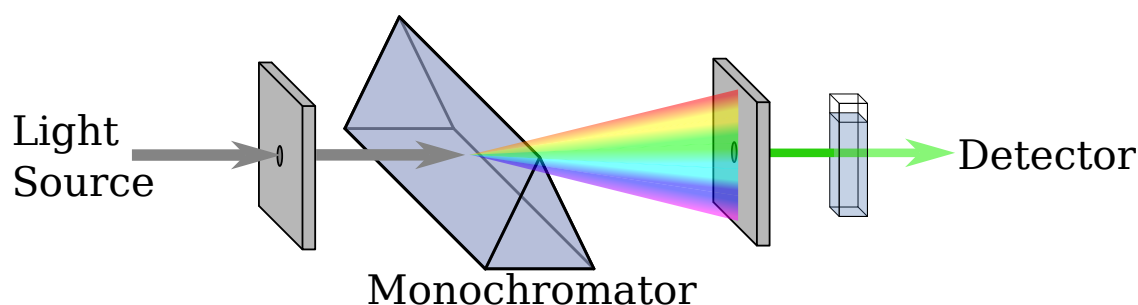


Figure 1.4: Block diagram of the fundamental components of a single-beam absorption spectrophotometer.

The light source must emit radiation across the entire wavelength range of interest with sufficient and stable intensity. For UV-visible spectrophotometry, two types of lamps are commonly used in commercial instruments: a deuterium lamp, which provides continuous radiation in the UV region (190–400 nm), and a tungsten-halogen lamp, which covers the visible and near-infrared region (350–900 nm). Many instruments automatically switch between the two lamps as the wavelength scan crosses the boundary between the UV and visible regions. In lower-cost and educational instruments, white LEDs are a popular alternative, as they are inexpensive, energy-efficient, and emit across much of the visible range, though with a non-uniform spectral output [1][4].

The wavelength selector is the component responsible for isolating a narrow band of wavelengths from the broadband output of the light source. In modern spectrophotometers, this is almost always achieved using a diffraction grating (a flat optical surface ruled with a large number of closely spaced parallel grooves). The diffraction grating spatially separates the incident light, directing different wavelengths into different angles according to the grating equation:

$$n\lambda = d \sin \theta \quad (4)$$

where n is the diffraction order (typically 1), λ is the wavelength of the diffracted light, d is the spacing between adjacent grooves, and θ is the diffraction angle. By rotating the grating or scanning the detector angle, different wavelengths can be directed onto the sample in sequence, allowing a full spectrum to be recorded. Older instruments used prisms as dispersive elements, but gratings have largely replaced them due to their more linear dispersion and lower cost [1].

The sample holder is typically a cuvette — a small transparent container with a precisely defined path length, usually 1 cm. Cuvettes for UV measurements must be made from materials that do not absorb UV radiation, such as quartz or fused silica. For measurements confined to the visible region, less expensive plastic or glass cuvettes are perfectly adequate [1].

The detector converts the transmitted light intensity into an electrical signal. The most common detector type in modern instruments is the photodiode, which generates a current proportional to the incident light intensity. In instruments designed to record a full spectrum simultaneously, an array of photodiodes (known as a photodiode array or PDA) can be used, replacing the need to scan through wavelengths mechanically. Photomultiplier tubes (PMTs), which offer much higher sensitivity, are used in high-end research instruments. For educational and low-cost devices, simpler detectors such as light-dependent resistors (LDRs) or phototransistors are also viable options [1][4].

Finally, the readout system processes the electrical signal from the detector and presents the result to the user. In modern instruments this is handled entirely by software running on a connected

computer, which can display the spectrum, perform baseline corrections, and carry out quantitative calculations automatically. In simpler home-built instruments, the detector signal is often read directly with a digital multimeter and the absorbance calculated by hand or with a simple script [1][8].

It is worth noting the distinction between single-beam and double-beam instrument designs. In a single-beam spectrophotometer, the sample and the blank (a reference solution, typically the pure solvent) are measured sequentially, and the ratio of their signals gives the transmittance. In a double-beam instrument, the light is split into two paths (one through the sample and one through the reference) and both are measured simultaneously. The double-beam design is more stable against fluctuations in the light source intensity and is therefore preferred in research-grade instruments, although single-beam designs are perfectly adequate for many educational and routine analytical applications [1][2].

2 Review of Low-Cost / DIY Spectrophotometers

The development of low-cost and home-built spectrophotometers has been an active area of interest in the educational and scientific community for several decades. The motivations behind these efforts are clear: commercial instruments are often prohibitively expensive for schools and smaller institutions, and even when available, they offer little insight into the physical principles of the measurement. A wide variety of designs have been proposed over the years, differing in complexity, cost, target audience, and performance. This section reviews the most relevant examples from the literature, with particular attention to the design choices made and the lessons that can be drawn for the present work.

One of the earliest and most accessible designs is that described by Lema et al. [6], who built a spectrophotometer from components readily available to anyone: a halogen flashlight as the light source, a compact disc (CD) as the diffraction element, a light-dependent resistor (LDR) as the detector, and a standard digital multimeter for readout. The entire assembly was mounted in a cardboard box, with two simple slits used to collimate the diffracted light onto the detector. Despite its extreme simplicity, this device was capable of producing recognizable absorption spectra and calibration curves for several biochemically relevant substances, including chlorophyll, starch, and proteins. The authors demonstrated its use in a series of teaching experiments in a biosciences course, showing that meaningful quantitative results could be obtained even with such minimal means. The main limitation of this design is its poor wavelength resolution (rotating the CD by a single degree corresponds to a wavelength change of approximately 25 nm) and the fact that it provides resistance readings rather than direct absorbance values. Nevertheless, it remains a compelling demonstration of how far ingenuity can go in replacing expensive equipment [6].

A more structured approach was taken by Knagge and Raftery [7], who designed a LEGO-based spectrophotometer for use in a graduate-level chemical instrumentation course at Purdue University. Their instrument incorporated a miniature filament lamp as the light source, a convex lens and an infrared filter for light conditioning, a 1000 lines/mm transmission diffraction grating, a spring-loaded rotating mirror for wavelength selection, a silicon photodetector, and an amplifier circuit connected to a digital multimeter. The use of LEGO bricks as optical mounts was motivated by their precise manufacturing tolerances, modularity, and mechanical stability, which make them well suited to holding optical components in alignment. The instrument achieved a wavelength resolution of approximately 8 nm and was used to record absorption spectra and calibration curves for potassium permanganate. Compared to a commercial Spectronic 20 instrument, the home-built

device showed higher stray light levels and a higher limit of detection, but its calibration curves displayed similar linearity. The total cost was approximately 200 USD. The authors emphasized that the educational value of the device lay not only in its use, but in the process of construction itself, through which students gained practical experience with the optical and electronic modules that make up any analytical instrument [7].

Building on the LEGO concept but targeting a much younger and less specialized audience, Albert, Todt, and Davis [8] developed a significantly simpler and cheaper LEGO spectrophotometer designed for use in high school physics and chemistry classes. Their design used a white LED as the light source, a single lens for collimation, a 1000 lines/mm slide-mounted diffraction grating, and a photodiode mounted on a rotatable arm for wavelength selection, with a digital multimeter as the readout. The entire apparatus was built on a standard LEGO baseplate and could be assembled and optimized in approximately 45 minutes. The total cost was around 25 USD, roughly one eighth of the cost of the Knagge and Raftery design. Despite its simplicity, the device produced absorbance measurements in close agreement with those from a commercial Ocean Optics USB2000+ spectrophotometer over the typical working range of absorbance values up to about 1, and was capable of recording full visible absorption spectra between approximately 420 and 660 nm. The authors noted that the use of familiar toy materials made the activity engaging for students, and that the transparent design effectively exposed the physical principles that are hidden in commercial black-box instruments [8].

A very different approach is represented by the work of Wang et al. [9], who demonstrated a smartphone-based spectrometer for colorimetric biosensing applications. Their device used the built-in LED flash of a smartphone as the light source, a compact disc fragment as the diffraction grating, and the smartphone's CMOS camera as the detector. The entire optical assembly was compact and required no external components beyond a simple sample cell. The smartphone camera captured the dispersed spectrum as an image, which was then processed using a dedicated Android application to extract the absorbance spectrum. The device was tested for the detection of glucose using an enzymatic colorimetric assay and for the detection of cardiac troponin I using peptide-functionalized gold nanoparticles, achieving limits of detection comparable to those of commercial bench-top instruments. While this design is more oriented toward portable point-of-care diagnostics than classroom education, it illustrates the remarkable potential of consumer electronics as spectroscopic platforms, and it highlights the growing role of smartphones and digital cameras as detectors in low-cost spectrophotometry [9].

González-Morales et al. [3] presented a portable low-cost UV-Vis spectrophotometer specifically designed for field measurements of environmental contaminants, with a particular application

to the detection of mercury ions in water using a synthesized chemosensor. Their instrument operated in the visible range from 400 to 700 nm and used a 1 W white LED as the light source, a DVD-derived diffraction grating mounted on a stepper motor for automated wavelength scanning, and a TEMT6000 phototransistor as the detector. The entire device was housed in a 3D-printed black enclosure and controlled by an Arduino microcontroller, which performed the wavelength sweep, acquired the detector signal, computed the absorbance, and transmitted the data to a computer. Two independent sample chambers allowed simultaneous measurement of the sample and a reference, improving stability. The total cost of the device was below 86 USD. Absorbance measurements agreed with those from a commercial Cary 60 spectrophotometer with error percentages below 10% for most concentrations, and below 5% for the mercury detection experiments. This design is particularly noteworthy for its level of integration and automation, demonstrating that a microcontroller-based instrument can approach commercial performance at a small fraction of the cost [3].

Finally, the Viking spectrophotometer developed by Kolesnichenko et al. [5] represents one of the most pedagogically sophisticated examples in the recent literature. Rather than aiming for maximum analytical performance, this design prioritized the learning experience above all else. The instrument was built from wooden holders, inexpensive magnifying glass lenses, a white LED light source with a replaceable aluminum foil slit, and a transmission diffraction grating mounted in a slide frame. Crucially, no fixed housing was used, all components were freely positioned on a flat table, allowing students to experiment with different configurations, observe the effects of misalignment, and develop genuine physical intuition. The spectrum was projected onto a white cardboard screen and photographed with a smartphone camera for digitization. The device supported both absorption and fluorescence measurements and was successfully deployed across a wide range of educational levels, from high school outreach activities to Master and PhD level courses at Lund University. Its total material cost was approximately 14 euros per set. The authors reported that engagement with the Viking spectrophotometer measurably improved the quality of students' laboratory reports and their confidence when subsequently operating professional instruments [5].

Taken together, these examples illustrate that low cost and high educational value are not mutually exclusive, and that the integration of a microcontroller with dedicated software can dramatically extend the functionality of a home-built device without greatly increasing its cost. These considerations have directly shaped the design decisions described in the following chapter.

3 Design and Implementation of the Spectrophotometer

3.1 Hardware

3.1.1 Idea

The starting point for the hardware design was the desire to build a functional absorption spectrophotometer that was as simple and inexpensive as possible, while remaining open to future refinement and improvement. A key decision in this direction was the choice of the SparkFun [AS7262](#) Visible Spectral Sensor as the core detection element. This sensor measures light intensity in six fixed channels across the visible spectrum, centered at 450, 500, 550, 570, 600, and 650 nm, and communicates digitally with a microcontroller via I2C. Its main advantage for this application is that it eliminates the need for a diffraction grating and a rotating wavelength selector entirely, the spectral discrimination is handled internally by the sensor itself. This greatly simplifies the optical layout and reduces the number of precision components required, at the cost of spectral resolution, since the instrument can only sample six discrete wavelengths rather than a continuous spectrum.

As the microcontroller, an [Arduino Uno](#) was chosen. This is a natural pairing with the AS7262 sensor, as the Arduino provides the I2C interface needed to communicate with it, can supply the sensor with power, and can transfer the acquired data to a PC via its USB connection. The Arduino Uno is also widely available, inexpensive, and well documented, making it a suitable platform for an educational instrument.

For the light source, an incandescent bulb was preferred over an LED. While LEDs are cheaper and more energy efficient, their emission spectrum is strongly peaked at specific wavelengths, which can result in uneven illumination across the six detection channels of the sensor. An incandescent bulb, by contrast, emits a broad and relatively smooth blackbody-like spectrum across the entire visible range, providing more uniform illumination of all channels. The chosen bulb is a Goobay product with an E10 socket, rated at 3.5 V and 0.2 A, giving a power consumption of 0.7 W. This low current draw is important, as the 5V power pin of the Arduino Uno can supply a limited current, and exceeding it risks damaging the board.

The optical layout follows a straightforward single-beam transmission configuration. Light from the bulb passes through a vertical slit, which collimates it into a narrow beam. This beam then passes through the cuvette, which holds the sample solution. A second slit placed after the cuvette

further restricts the beam before it reaches the AS7262 sensor. Standard commercial cuvettes with a 10×10 mm inner cross section and 1 cm path length were used, consistent with the standard path length assumed in Beer-Lambert law calculations. The structural frame holding all of these components together was designed in 3D modeling software and printed using a high-precision 3D printer with PLA filament, resulting in a compact enclosure slightly smaller than a 6×10×25 cm block. The detailed plans (and software files) can be found in the GitHub repository: https://github.com/bacstamas/Spectrometer_2026.

3.1.2 Obstacles

The development of the hardware went through two main iterations, driven by a series of practical problems encountered during testing.

The first version of the frame was modular in design, meaning that the positions of the light source, the slits, and the sensor could be adjusted along the optical axis. This was useful in the early stages of development, as it allowed the optimal spacing between components to be found experimentally. However, this first version had no cover or enclosure around the optical path, meaning that ambient light from the room could reach the sensor and contaminate the measurements. As a result, all measurements with this version had to be performed in a darkened room, which was clearly impractical for regular use.

A further issue that emerged during testing was the need to control the intensity of the light source. The AS7262 sensor has a fixed integration time and gain setting for each measurement, and if the light reaching it is too intense, the sensor saturates and the readings become unreliable. A way to reduce the light intensity without modifying the optical layout was therefore needed.

Finally, the PLA filament used for 3D printing presented an unexpected problem. The filament available was blue in color, and it turned out to be partially translucent to ambient light, allowing a non-negligible amount of stray light to penetrate the walls of the enclosure even in the second, closed version of the frame.

3.1.3 Implementation

The second version of the frame addressed the problems identified during testing with the first. All components (the light source, the two slits, the cuvette holder, and the sensor) were given fixed, precisely dimensioned positions inside a closed enclosure, based on the optimal spacing determined during the first phase. The enclosure included a removable cover to allow access to the cuvette holder for changing samples, while keeping the rest of the optical path shielded

from ambient light. A dedicated bay was also added to securely mount the Arduino Uno board, positioned with the USB Type-B connector accessible from the outside of the enclosure, so that the connection to the PC could be made without opening the instrument. Crucially, the Arduino was physically separated from the optical components inside the box, so that the indicator LEDs on the board would not interfere with the sensor readings. The internal structure of the assembled spectrometer is shown in Figure 3.2.

To solve the problem of stray light penetrating through the blue PLA walls, the exterior and interior surfaces of the enclosure were painted with a matte black paint of the type used industrially to produce surfaces with near-unity emissivity for thermal imaging applications. This paint effectively blocked the residual light transmission through the plastic walls and also minimized internal reflections within the enclosure.

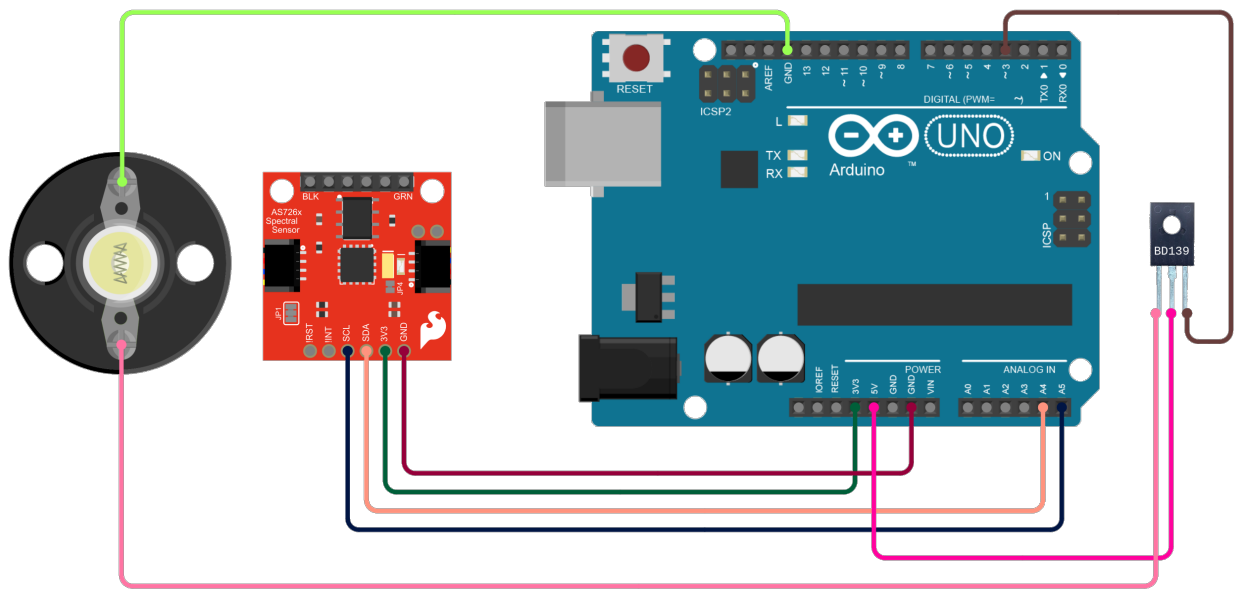


Figure 3.1: Circuit diagram of the spectrophotometer electronics, showing the connections between the Arduino Uno, the SparkFun AS7262 spectral sensor (I2C interface), the BD139 NPN transistor, and the incandescent light bulb. The transistor is driven by the PWM output on pin 3 of the Arduino and controls the current through the light source.

The light intensity control problem was solved by adding a [BD139](#) NPN bipolar transistor to the circuit. The transistor was wired with its collector connected to the Arduino's 5V supply pin, its emitter connected to the positive terminal of the light bulb, and the negative terminal of the bulb connected to the Arduino's ground. The base of the transistor was driven by pin 3 of the Arduino, which is a PWM-capable output. By varying the duty cycle of the PWM signal, the effective

voltage across the bulb can be adjusted continuously, allowing the light intensity to be tuned to a level that keeps the sensor within its linear operating range without saturating it. This simple addition significantly improved the reliability and reproducibility of the measurements. See Figure 3.1 for the circuit diagram.

Several sub-components, including the sensor holder, the cuvette holder, and the box cover, went through additional minor refinements beyond the two main frame iterations, as small geometric adjustments were needed to improve the fit and reduce stray light leakage around the edges. The final assembled instrument is a compact, self-contained device that can be connected to any computer via a standard USB cable and operated without any need for a darkened environment.

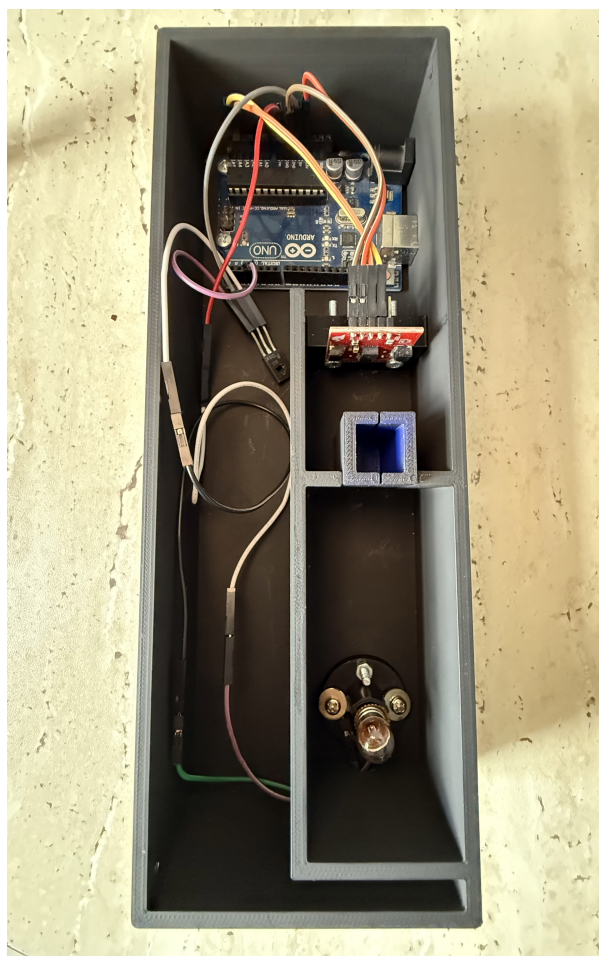


Figure 3.2: Photo of the inside of the spectrometer.

3.2 Software

The software component of the spectrophotometer system consists of two separate programs: the firmware running on the Arduino microcontroller, which is responsible for directly operating the

sensor and acquiring data, and a Java desktop application running on the PC, which serves as the interface between the user and the instrument. The two communicate through a serial connection over USB.

The **Arduino firmware** is intentionally kept simple and lightweight. On startup, it initializes the AS7262 sensor and sets a default configuration for the measurement parameters. It then enters a waiting loop, listening for commands arriving through the serial port.

When a measurement is requested, the command arrives as a formatted string specifying all relevant parameters: integration time, gain, number of averages, measurement mode, and light intensity. The firmware parses this string, applies the specified settings to the sensor, and adjusts the PWM signal on pin 3 to set the light source intensity accordingly. It then performs the measurement the requested number of times and computes the average across repetitions for each of the six spectral channels. The result is sent back to the PC as a single comma-separated line of six floating point values, one per channel.

The AS7262 sensor library supports two measurement modes: a raw mode, which returns the unprocessed digital counts from each photodetector channel, and a calibrated mode, which applies internal corrections to return values in units of calibrated irradiance. Both modes are accessible through the firmware and can be selected by the user from the PC software.

The **PC-side software** is written in Java, a choice motivated by several practical considerations. Java is inherently cross-platform, the same application can run on Linux, Windows, and macOS without significant modification, and it lends itself naturally to well-structured, object-oriented design that can be extended and maintained over time. The application was developed and tested on Linux but is intended to be portable across operating systems. It uses the *jSerialComm* library for serial port communication, the *XChart* library for data visualization and the *Jackson* library for handling JSON files.

The architecture of the application is organized around a small set of well-defined classes, each with a clear responsibility. This separation of concerns makes the codebase easy to navigate and extend without requiring a complete redesign.

3.2.1 Structure

The *Spectrometer* class is the core of the application. It is responsible for all communication with the Arduino hardware. On instantiation, it automatically scans the available serial ports on the system (using the *jSerialComm* library) and identifies the connected Arduino by checking port descriptors and system names. Once connected, it configures the serial port parameters (baud rate,

data bits, stop bits, parity) opens the connection, and flushes any startup messages sent by the Arduino. It exposes methods to configure the measurement parameters and to trigger measurements, handling the sending of the formatted command string and the reception and parsing of the response. Each call to the measurement method can request multiple spectra in sequence, each acquired with on-board averaging by the Arduino.

The *MeasurementSet* class acts as a container for all data associated with a single acquisition session. It stores the list of individual spectra recorded during that session, the full set of measurement parameters used, and a unique name composed of a user-defined label and the date and time of acquisition. It provides methods for computing the mean and standard deviation across the recorded spectra for each channel, for formatting this information as a readable string, and for saving the entire measurement set to a JSON file or loading it back from one. JSON format was chosen because it is human-readable and easy to handle.

The *Visualizer* class handles all data visualization. It takes a *MeasurementSet* as input and can produce two types of charts: a spectrum plot showing the intensity values across the six channels of the sensor, and an absorption spectrum computed from two separate measurement sets representing the reference and the sample. The absorbance at each channel is calculated according to the Beer-Lambert law as:

$$A = -\log_{10} \left(\frac{I_{sample}}{I_{reference}} \right) \quad (5)$$

where I_{sample} and $I_{reference}$ are the mean intensities of the sample and reference measurements at each channel respectively. The *Visualizer* class uses the *XChart* library to produce the actual chart objects, which are then embedded in the GUI.

The remaining classes handle the graphical user interface. The *Main* class serves as the application entry point, launching the GUI on the appropriate thread. The *MainWindow* class implements the main application window, while the remaining classes manage the panels, menus, and dialogs, and handle specific user interactions.

3.2.2 Features

The main window of the application is organized around a central list that displays all measurement sets currently loaded in memory, each identified by its unique name. From the main window, the user can connect to the spectrometer and configure the measurement parameters through a dedicated dialog, which provides input fields for integration time, gain, number of averages, number of measurements, measurement mode, and light intensity. Once configured, a measurement can be triggered with a single button press. The acquired measurement set is automatically added to the

list and displayed in a text area showing its parameters and statistics.

Measurement sets can be saved to and loaded from plain text files, allowing results to be stored for later analysis or shared between sessions. The user can also load previously saved measurements without connecting to the hardware, making it possible to perform data analysis and visualization independently of the instrument.

For visualization, two separate plot types are available. The spectrum plot displays the raw or calibrated intensity values across the six sensor channels for a selected measurement set, while the absorption plot computes and displays the absorbance spectrum from a user-selected reference and sample pair. Both plot types offer configurable options accessible through dialog windows: the x-axis can be displayed in wavelength (nm) or in frequency (THz), the y-axis values can be shown as raw values or normalized to the maximum, the plot style can be switched between a line curve and a bar chart, and error bars representing the standard deviation across repeated measurements can be toggled on or off.

3.2.3 Data Processing and Calibration

The data processing performed by the software is deliberately straightforward, reflecting the educational purpose of the instrument. The primary computation is the averaging of repeated spectra, which is handled partly by the Arduino firmware, which averages multiple sensor readings within a single acquisition, and partly by the *MeasurementSet* class, which computes the mean and standard deviation across multiple sequential acquisitions requested by the user. This two-level averaging scheme gives the user flexibility in balancing measurement speed against statistical reliability.

The absorbance calculation follows directly from the Beer-Lambert law, as described above. No additional baseline correction or spectral smoothing is applied, keeping the processing chain transparent and easy to follow.

3.3 Experimental Results and Validation

With the hardware and software components completed, a series of experiments was carried out to characterize the performance of the instrument and to validate its suitability for quantitative absorbance measurements.

3.3.1 Stray Light Test

The first test performed was aimed at verifying that the closed enclosure of the spectrophotometer effectively excludes ambient light. An empty cuvette was placed in the cuvette holder, the cover of the box was closed, and the light source intensity was set to 0, meaning the bulb was switched off via the PWM control. Under these conditions, repeated measurements returned a value of exactly zero on each of the six channels of the sensor. This result confirms that no stray or ambient light reaches the detector when the light source is off, validating the effectiveness of the matte black coating applied to the interior of the enclosure and ruling out the presence of any light leaks through the housing.

3.3.2 Reproducibility Test

To assess the reproducibility of the measurements, a solution of yellow food coloring at a concentration of 2.5% was prepared and measured repeatedly. Ten independent measurement sets were acquired, each consisting of 100 individual measurements per channel (integration time: 100 ms, gain: 16, averaging: 1, mode: raw, light intensity: 100). All ten sets were plotted together, as shown in Figure 3.3, where each of the six channels is represented by ten overlapping traces of 100 points each.

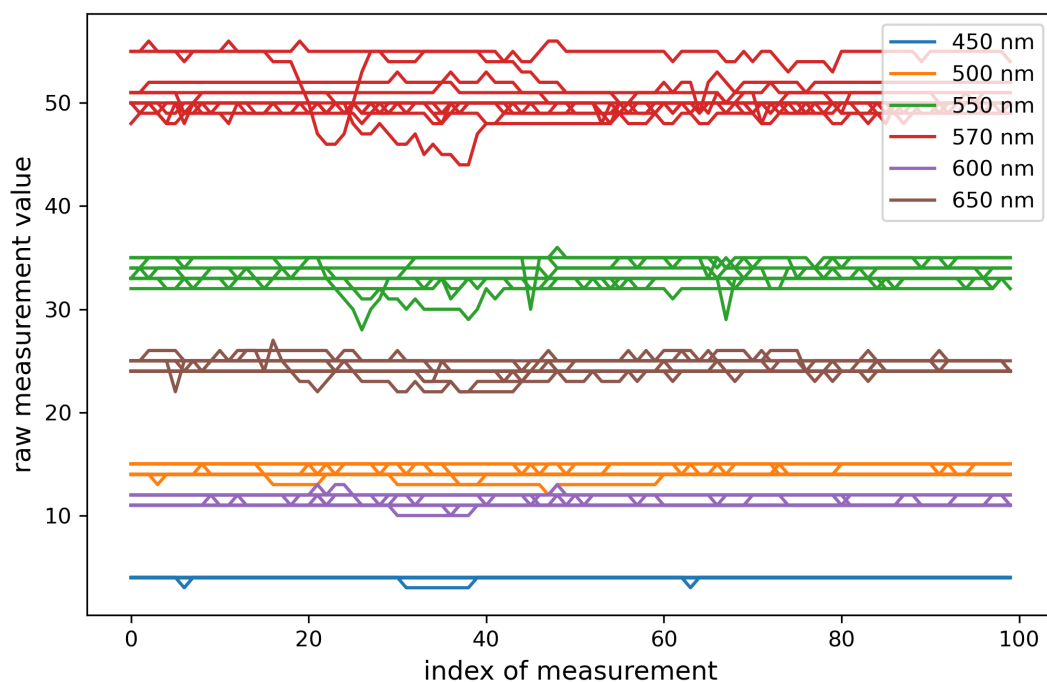


Figure 3.3: Raw measurement values recorded across the six channels of the AS7262 sensor (450, 500, 550, 570, 600, and 650 nm) for ten independently acquired measurement sets of a 2.5% yellow food coloring solution, each consisting of 100 individual measurements.

The plot shows that the measured values remain clustered around a stable baseline for each channel, with some random scatter present throughout the dataset. Averaging across all 1000 individual measurements per channel (10 sets $\times$ 100 measurements), the mean raw values obtained were $\{3.99, 14.251, 33.428, 50.629, 11.336, 24.334\}$ for the 450, 500, 550, 570, 600, and 650 nm channels respectively, with corresponding standard deviations of $\{0.099, 0.522, 1.101, 2.004, 0.501, 0.696\}$. The relative standard deviation remains below approximately 4% for every channel, which indicates a satisfactory level of reproducibility for an instrument of this cost and complexity. This noise may be attributable to small imperfections such as scratches on the cuvette surface and slight variations in its exact positioning within the holder between insertions.

3.3.3 Long-Term Stability Test

A further test was conducted to check whether the solution itself remained stable over time, an important consideration since sedimentation or settling of the food coloring could introduce a systematic drift in absorbance unrelated to the instrument's performance. The same yellow food coloring solution and the same measurement parameters used in the reproducibility test were employed. A

cuvette containing the solution was measured once, left undisturbed inside the instrument for 24 hours, and then measured again without being touched or repositioned in between. The difference between the two measurements was found to be smaller than the standard deviations already observed in the reproducibility test, indicating that no significant settling or degradation of the solution occurred over this timescale, and that any short-term drift in the instrument's readings is dominated by the handling-related noise identified above rather than by sample instability.

3.3.4 Calibration Curves

Finally, the linearity of the instrument's response was tested by preparing a series of yellow food coloring solutions at concentrations ranging from 0.8% to 10%, and measuring the absorbance of each at all six sensor channels. The resulting calibration curves are shown in Figure 3.4 (one subplot for each channel) together with a linear best-fit line.

Across all six channels, the measured absorbance values follow the linear trend predicted by the Beer-Lambert law reasonably well, particularly in the central concentration range. Some scatter around the fitted line is visible, which is consistent with the absolute measurement noise identified in the reproducibility test. The channels at 550, 570, and 600 nm, which are closer to the absorption maximum of the yellow dye, display the clearest linear behavior and the largest overall absorbance range, while the 450 nm channel, which lies further from the absorption peak, shows comparatively more scatter relative to its smaller dynamic range. Overall, these results confirm that the constructed instrument is capable of producing calibration curves with a clearly recognizable linear character across the entire visible range covered by the sensor, supporting its intended use for quantitative concentration measurements in an educational context.

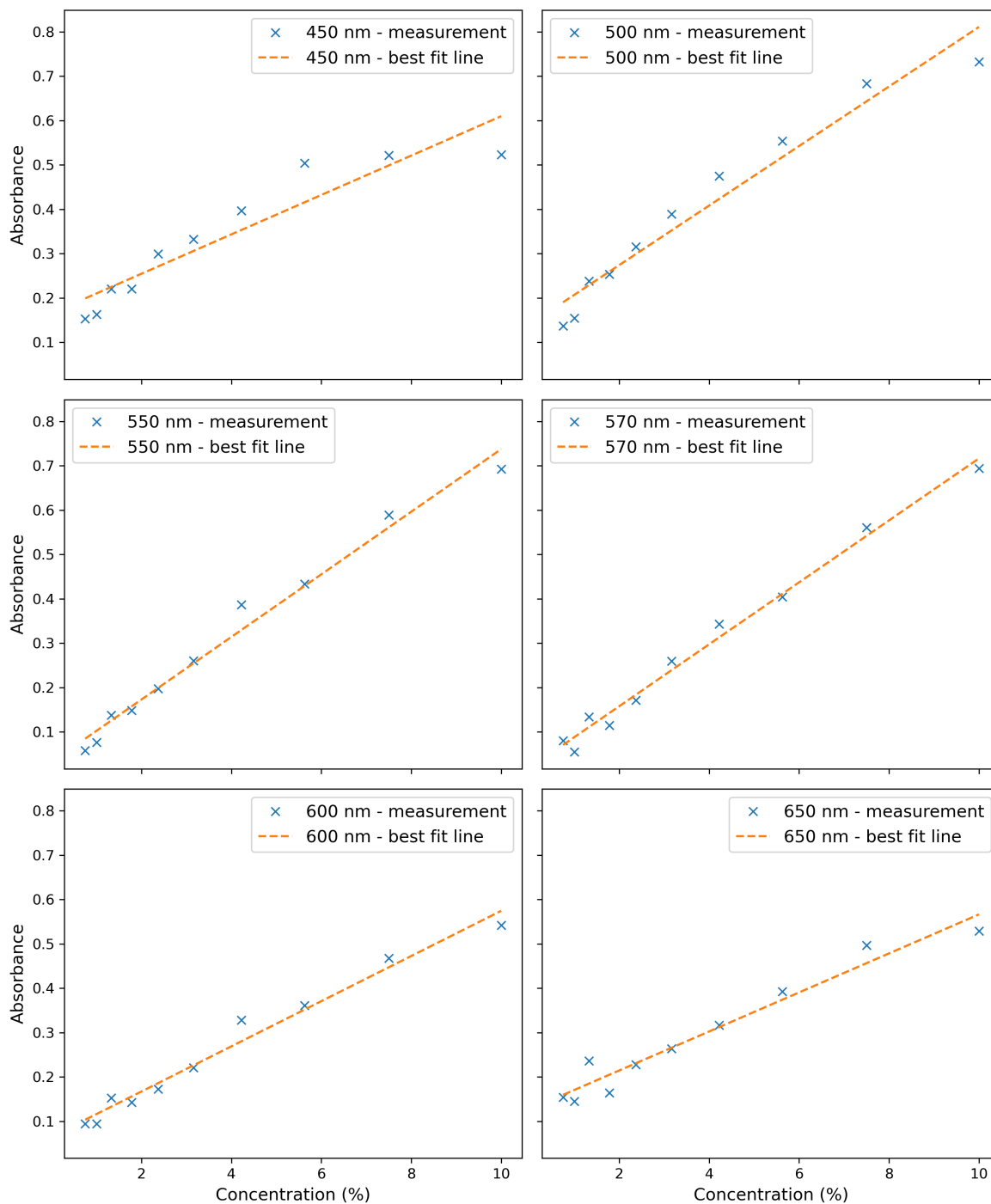


Figure 3.4: Six calibration curves showing absorbance against concentration (%), along with their corresponding best-fit lines, at each of the sensor's wavelengths.

Conclusion

The primary objective of this thesis was the design, construction, and validation of a low-cost absorption spectrophotometer intended for educational use. The developed instrument combines simple hardware with dedicated software for data acquisition, visualization, and absorbance calculation. Experimental testing demonstrated that the spectrometer is capable of producing reproducible measurements and calibration curves that follow the linear behavior predicted by the Beer-Lambert law. While the measurements exhibit a non-negligible level of noise, the results show that the instrument is sufficiently reliable for educational and demonstration purposes, provided that appropriate care is taken during sample preparation and measurement.

One of the main strengths of the instrument is its combination of low cost and transparency. Unlike commercial spectrophotometers, whose internal operation is often hidden from the user, the present design exposes the essential elements of an absorption measurement system and makes the underlying physical principles easier to understand. As a result, the spectrometer is well suited for use in secondary-school and undergraduate laboratory activities. Students can gain practical experience with solution preparation, reference measurements, absorbance calculations, the construction of calibration curves, and the quantitative determination of concentrations. More advanced educational projects could even include the determination of the molar absorptivity of a substance through Beer-Lambert analysis.

Beyond its use as a measurement instrument, the reproduction of the spectrometer itself could serve as a valuable educational project for physics students at the BSc or MSc level. Constructing the device requires engagement with several important areas of modern scientific and engineering practice, including absorption spectroscopy, 3D design and printing, microcontroller programming, object-oriented software development, and experimental testing. The process of building and validating the instrument provides a deeper understanding of both the physical principles and the practical challenges involved in spectroscopic measurements.

Although the instrument successfully fulfills its intended educational purpose, several opportunities for further development remain. One possible improvement would be the replacement of the AS7262 sensor with a sensor offering a larger number of spectral channels, increasing the spectral resolution of the measurements. Another possibility would be to adopt a more traditional spectrometer architecture based on a rotatable diffraction grating and a simpler photodetector, allowing the acquisition of a continuous spectrum rather than measurements at six discrete wavelengths. Such modifications could improve the analytical capabilities of the instrument while preserving its

educational value.

A particularly promising direction for future development would be the replacement of the Arduino Uno with an ESP32 microcontroller. The additional processing power and built-in Wi-Fi connectivity of the ESP32 would make it possible to implement the user interface as a web application hosted directly by the instrument. In such a design, users could access the spectrometer through any device equipped with a web browser, without the need to install dedicated software. This would simplify deployment in classroom environments, reduce platform-dependence, and further increase the accessibility of the system.

In conclusion, the developed spectrophotometer demonstrates that quantitative absorption measurements can be achieved with inexpensive and widely available components. Although it cannot compete with commercial instruments in terms of precision or spectral resolution, it successfully fulfills its intended role as an educational tool and provides a practical platform for teaching both spectroscopy and scientific instrumentation.

References

- [1] OWEN, Tony. *Fundamentals of UV-visible spectroscopy*. 2000.
- [2] Demystifying spectroscopic analysis through expert insights and practical resources: www.spectroanalysis.com
- [3] GONZÁLEZ-MORALES, David, et al. *Development of a low-cost UV-Vis spectrophotometer and its application for the detection of mercuric ions assisted by chemosensors*. *Sensors*, 2020, 20.3: 906.
- [4] KOVARIK, Michelle L.; CLAPIS, Julia R.; ROMANO-PRINGLE, K. Ana. *Review of student-built spectroscopy instrumentation projects*. *Journal of Chemical Education*, 2020, 97.8: 2185-2195.
- [5] KOLESNICHENKO, Pavel V., et al. *Viking spectrophotometer: a home-built, simple, and cost-efficient absorption and fluorescence spectrophotometer for education in chemistry*. *Journal of Chemical Education*, 2023, 100.3: 1128-1137.
- [6] LEMA, Martín A.; ALJINOVIC, Ernesto M.; LOZANO, Mario E. *Using a homemade spectrophotometer in teaching biosciences*. *Biochemistry and Molecular Biology Education*, 2002, 30.2: 106-110.
- [7] KNAGGE, Kevin; RAFTERY, Daniel. *Construction and evaluation of a LEGO spectrophotometer for student use*. *The Chemical Educator*, 2002, 7.6: 371-375.
- [8] ALBERT, Daniel R.; TODT, Michael A.; DAVIS, H. Floyd. *A low-cost quantitative absorption spectrophotometer*. *Journal of Chemical Education*, 2012, 89.11: 1432-1435.
- [9] WANG, Yi, et al. *Smartphone spectrometer for colorimetric biosensing*. *Analyst*, 2016, 141.11: 3233-3238.
- [10] Advancing analytical chemistry knowledge through expert insights and research.: www.analchemsci.com